Daniele Mancinelli

Curriculum Vitae

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Milano, Italy
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General Information

First name	Daniele
Last name	Mancinelli
Place and date of birth	Rome (Italy), 28 August 1996
Citizenship	Italian
ORCID	https://orcid.org/0000-0003-3854-2414
	Google Scholar Profile

Current position

30/10/2026 - present **Researcher RTT at Department of Economics and Finance, University of Rome Tor Vergata (Italy)**
GSD: 13/STAT-04, SSD: STAT-04/A, METODI MATEMATICI DELL'ECONOMIA E DELLE SCIENZE ATTUARIALI E FINANZIARIE

Past academic positions

1/12/2025 - 29/10/2026 **Researcher RTDA at Department of Mathematics, Politecnico di Milano (Italy)**
GSD: 13/STAT-04, SSD: STAT-04/A, METODI MATEMATICI DELL'ECONOMIA E DELLE SCIENZE ATTUARIALI E FINANZIARIE
Research project: *Pricing, Protection, and Allocation: New Frontiers in Portfolio Strategy Design*

1/2/2024 - 30/11/2025 **Post-doctoral fellow at Department of Economics and Finance, University of Rome Tor Vergata (Italy)**
GSD: 13/STAT-04, SSD: STAT-04/A, METODI MATEMATICI DELL'ECONOMIA E DELLE SCIENZE ATTUARIALI E FINANZIARIE
Research project: *Investment strategies for green transition and emission reduction*
Advisor: Prof. Katia Colaneri.

Education

1/11/2020-30/1/2024 **Ph.D. in MODELS FOR ECONOMICS AND FINANCE**
Department of Methods and Models for Territory, Economics and Finance (MEMOTEF), Sapienza, University of Rome
Grade: Excellent
Thesis: *New Insights on Portfolio Insurance Strategies for financial and longevity risk management*
Supervisor: Prof. Immacolata Oliva.

2018 - 2020 **Master in Finance and Insurance (Laurea Magistrale)**
Sapienza, University of Rome
Grade: 110/110 *cum laude*
Thesis: *Analisi delle strategie di Portfolio Insurance*
Supervisor: Prof. Immacolata Oliva.

2015 - 2018 **Bachelor in Economics and Finance (Laurea Triennale)**
Università degli Studi Roma 3
Grade: 110/110 *cum laude*
Thesis: *Utilità attesa e scelte di portafoglio in Tobin*
Supervisor: Prof. Enrico Sergio Levrero.

Teaching and Academic Service

Teaching Experience

2026 Teaching assistant, *Fondamenti di Matematica*, Politecnico di Milano.
2026 Teaching assistant, *Insurance and Econometrics*, Politecnico di Milano.
2025 Teaching assistant, *Calculus I*, LUISS Guido Carli.
2025 Teaching assistant, *Metodi Matematici per la Finanza*, LUISS Guido Carli.
2024 Teaching assistant, *Quantitative Models for Data Science*, LUISS Guido Carli.
2024 Teaching assistant, *Metodi Matematici per la Finanza*, LUISS Guido Carli.
2023 Teaching assistant, *Quantitative Models for Data Science*, LUISS Guido Carli.
2023 Teaching assistant, *Matematica generale*, Sapienza University of Rome.
2022 Teaching assistant, *Matematica generale*, Sapienza University of Rome.
2022 Teaching assistant, *Econometria Finanziaria*, LUISS Guido Carli.
2021 Teaching assistant, *Matematica generale*, Sapienza University of Rome.

Discussant for Bachelor's and Master's Theses

2026 *Credit Value Adjustment in Reinsurance: A Quantitative Framework and Case Study*, Master's thesis, Enrico Baggio, Politecnico di Milano. Supervisor: Prof. Alessandro Calvia.
2026 *Optimal Abatement Investment under Deterministic, Pure-Jump, and Mean-Reverting Carbon Tax Dynamics*, Master's thesis, Carlotta Carzaniga, Politecnico di Milano. Supervisor: Prof. Michele Azzone.
2026 *Differential Prediction-Enhanced Monte Carlo: A Fast Machine Learning-Aided Variance Reduction Method*, Master's thesis, Luca Amadori, Politecnico di Milano. Supervisor: Prof. Daniele Marazzina.

Funding Information

2024 Participation as member of the research unit to the research project *Do you really want to Earth me: green washing mitigation and firms behaviour*, Progetto di Ateneo 2024, University of Rome Tor Vergata. Funded amount: 7591,50.
2024 Participation as member of the research unit to the research project *Facts and Theory about the role of Gender Representation: the case of Italian Academic Careers*, Progetto di Ateneo 2024, University of Rome Tor Vergata. Funded amount: 11119,50 euro.
2023 Participation as Principal Investigator to the research project *Pension fund with longevity risk: an optimal portfolio insurance approach*, Avvio alla ricerca 2023 Sapienza Università di Roma. Funded amount: 2000 euro.
2023 Participation as member of the research unit to the research project *Stochastic Optimization Problems in Insurance, Finance and Economics*, Progetto di Ateneo 2023, Sapienza Università di Roma. Funded amount: 32650 euro.
2022 Participation as member of the research unit to the research project *Frictions meet incompleteness: no-arbitrage solutions for derivative pricing and investment opportunities,* Progetto di Ateneo 2022 Sapienza Università di Roma. Funded amount: 10000 euro.

2021 Participation as member of the research unit to the research project *Facing emerging risks: an actuarial perspective,* Progetto di Ateneo 2021 Sapienza Università di Roma. Funded amount: 12000 euro.

Society Memberships

2025-present Member of Association for Mathematics Applied to Social and Economic Sciences (A.M.A.S.E.S.)

2026-present Member of the Bachelier Finance Society

Awards

2025 **Best Junior Paper Award**, A.M.A.S.E.S. – Association for Mathematics Applied to Social and Economic Sciences, XLIX Annual Conference (Firenze, 11–13 Sep 2025), for the paper *Design and hedging of unit-linked life insurance with environmental factors*.

Language Skills

Italian mother tongue

English fluent

Computer Skills

Program Languages Matlab, R, Python, VBA, C++.

Software system Mathematica, LaTeX.

Application software Microsoft Word, Excel, PowerPoint, Outlook.

Research

Research interest Main topics include:

- Portfolio optimization, Pension funds design, Portfolio insurance, Quantitative methods for Green Finance
- Numerical methods for PDEs and PIDEs, Monte Carlo methods
- Stochastic Analysis, Stochastic Control and Games

Summer School

2021 **SoFiE Financial Econometrics Summer School: "The Econometrics of Derivatives Markets"**
Kellogg School of Management, Northwestern University.

2021 **Summer School: "From Network to Neural Network in Finance"**
Lake Como School of Advanced Studies.

2018 **Summer School: "Strumenti e tecniche Matlab per il calcolo parallelo, l'apprendimento automatico e l'analisi massiva dei dati"**
Università degli Studi di Palermo.

2018 **Summer School: "Programmazione e calcolo scientifico con Matlab"**
Università degli Studi di Palermo.

Journal Services

Co-Guest Editor of Special Issue *Financial Modeling with Frictions*, Frontiers in Applied Mathematics and Statistics.

Referee for: Mathematics and Financial Economics, Scandinavian Actuarial Journal.

Contributed Talks at International Conferences

XIII Bachelier World Congress, University of Bologna, Bologna (Italy), 29 June-3 July 2026. Title: *Carbon-Sensitive Fund Construction and Hedging for Green Unit-Linked Life Insurance*.

5th Italian Meeting on Probability and Mathematical Statistics, University of Palermo, Palermo (Italy), 8-12 June 2026. Title: *Optimal reinsurance with unobservable claim arrival intensity: a Stackelberg differential game*.

XLIX Annual Meeting of the Italian Association for Mathematics Applied to Economic and Social Sciences, University of Firenze, Toscana, Firenze (FI) (Italy), 11-13 September 2025 Title: *Design and hedging of unit-linked life insurance with environmental factors.*

Vienna Congress on Mathematical Finance (VCMF 2025), WU (Vienna University of Economics and Business), Vienna (Austria), 9-11 July 2025, Title: *Design and hedging of unit-linked life insurance with environmental factors.*

XXVI Quantitative Finance Workshop (QFW 2025), University of Palermo, Sicilia, Palermo (PA) (Italy), 15-17 April 2025. Title: *Optimal design of carbon-penalized proportional portfolio insurance strategies in a partially observable market model.*

XLVIII Annual Meeting of the Italian Association for Mathematics Applied to Economic and Social Sciences (AMASES), Università degli Studi di Napoli Federico II, Ischia (Italy), 5-7 September 2024. Title: *Optimal design of proportional portfolio insurance strategies with environmental, social, governance (ESG) factors.*

XLVII Annual Meeting of the Italian Association for Mathematics Applied to Economic and Social Sciences (AMASES), Università degli Studi di Milano-Bicocca, Milano (Italy), 20-22 September 2023. Title: Optimal design of a new class of Proportional portfolio insurance strategy in a jump-diffusion framework.

26th International Congress on Insurance: Mathematics and Economics (IME 2023), Heriot-Watt University, Edinburgh (UK), 4-7 July 2023. Title: *Optimal design of a new class of Proportional portfolio insurance strategy in a jump-diffusion framework.*

XXIV Quantitative Finance Workshop (QFW 2023), University of Cassino and Southern Lazio, Gaeta (LT) (Italy), 20-22 April 2023. Title: *Pension fund with longevity risk: an optimal portfolio insurance approach.*

XLVI Annual Meeting of the Italian Association for Mathematics Applied to Economic and Social Sciences (AMASES), University of Palermo, Sicilia, Palermo (PA) (Italy) 22-24 September 2022. Title: *Retirement capital accumulation in DC schemes in a jump diffusion framework with longevity risk.*

XXIII Quantitative Finance Workshop (QFW 2022), Tor Vergata University of Rome, Lazio, Roma (RM) (Italy), 31 March-1 April 2022. Title: *A performance analysis of portfolio strategies with guaranteed minimum equity exposure.*

International Risk Management Conference (IRMC 2021), University of Cagliari, Sardegna, Cagliari (CA) (Italy), 1-2 October 2021. Title: *A performance analysis of portfolio strategies with guaranteed minimum equity exposure.*

Seminars

Politecnico di Milano, Milano (MI) (Italy), 15 April 2025. Title: *Optimal design of carbon-penalized proportional portfolio insurance strategies in a partially observable market model.*

Scientific Event Organization (Conferences and Workshops)

Co-organizer of the session *Exploring Uncertainty in Financial Modelling and Quantitative Risk Management* at the 49th AMASES Meeting, September 2025.

Publications

A-class according to criterion for National habilitation GSD STAT/04

K. Colaneri, F. D'Amario, D. Mancinelli (2026) *Carbon-Penalised Portfolio Insurance Strategies in a Stochastic Factor Model with Partial Information*, Scandinavian Actuarial Journal. https://doi.org/10.1080/03461238.2026.2682367.

K. Colaneri, D. Mancinelli, I. Oliva (2025) *On the optimal design of a new class of proportional portfolio insurance strategy in a jump-diffusion framework*, Scandinavian Actuarial Journal. https://doi.org/10.1080/03461238.2025.2478443

M. Di Giacinto, D. Mancinelli, M. Marino, I. Oliva (2024) *Pension fund with longevity risk: an optimal portfolio insurance approach*, Insurance: Mathematics and Economics. https://doi.org/10.1016/j.insmatheco.2024.10.001

L. Di Persio, D. Mancinelli, I. Oliva, K. Wallbaum (2023) *Time-invariant portfolio strategies in structured products with guaranteed minimum equity exposure*, Applied Stochastic Models in Business and Industry. https://doi.org/10.1002/asmb.2805

Other publications

D. Mancinelli, A. Mazzon, I. Oliva, I. Stefani (2025) *Editorial: Financial modeling with frictions*, Frontiers in Applied Mathematics and Statistics. https://doi.org/10.3389/fams.2025.1641147

D. Mancinelli, I. Oliva (2023) *Constant or variable? A performance analysis among portfolio insurance strategies*, Risks. https://doi.org/10.3390/risks11060105

D. Mancinelli (2022) *Managing cash-in risk embedded in Portfolio Insurance strategies: a review*, ANNALI DEL DIPARTIMENTO DI METODI E MODELLI PER L'ECONOMIA, IL TERRITORIO E LA FINANZA, https://doi.org/10.13133/2611-6634/1414

Preprints

K. Colaneri, A. Cretarola, E. Lombardo, D. Mancinelli (2025) *Carbon-Sensitive Fund Construction and Hedging for Green Unit-Linked Life Insurance*, https://doi.org/10.48550/arXiv.2509.05676. Under review.

Work in Progress

A random utility model on the adoption of Central Bank Digital Currency, with E. Barucci, M. Tolotti

Multilevel Monte Carlo Valuation of Green Unit-Linked Life Insurance Policies, with K. Colaneri, E. Lombardo

A Stackelberg game between a primary insurer and a reinsurer with symmetric and asymmetric information, with A. Calvia, K. Colaneri, E. Lombardo

Optimal Design of Individual Pension Plans: A Stackelberg Differential Game Approach, with M. Aleandri, K. Colaneri, M. Di Giacinto

Milano, 2026-11-11